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Unlocking muscle performance with functional isometrics

Exploring the science, applications, and programming of one of **strength training's** most overlooked techniques.

The term functional isometrics refers to a training methodology that combines isometric and dynamic strength to improve athletic performance.

This approach, introduced by Bob Hoffman and developed by experts such as John Rusin, offers an alternative for those who want to develop strength, muscle mass, and mind-muscle connection in a targeted way.

Let's explore how to integrate them into training programmes, and the advantages and disadvantages of this technique in the sporting domain.

What are functional isometrics?

Functional isometrics are based on the use of isometric contractions (where the muscle generates tension without changing length) integrated with dynamic movements.

This technique focuses on a limited portion of the Range of Motion (ROM), often in the most difficult phases of an exercise, such as the 'low' part of a squat or the stall point on a flat bench press.

The approach involves the use of racks/benches with limiters, which

restrict movement to a particular range. Athletes press against the limiters with maximum force for a set period of time, preceded by a dynamic phase.

This method allows the load to be directed to specific points of the movement, making it particularly useful for improving weaknesses or limitations in complex exercises.

How functional isometrics work?

The efficacy of functional isometrics lies in their ability to:

Increase maximum strength: By focusing on precise positions of the ROM, athletes can overload the muscular system in a targeted way, allowing them to develop greater overall lifting capacity.

Improve neuromuscular coordination: Isometric contractions allow greater recruitment of motor units, improving the efficiency of muscle fibre activation.

Overcome staleness: By working on crucial phases of movement, specific weaknesses can be improved, making the entire exercise more fluid.

Increase muscle control: The combination of isometric and dynamic contractions strengthens the

mind-muscle connection, essential for athletes who require precision in movement.

Enable safer training: By focusing on limited portions of the ROM, the risk of compensatory movements or excessive loads across the full range of motion is reduced.

Training parameters

The integration of functional isometrics into training programmes requires careful planning and in-depth knowledge of the exercises involved. We analyse the key training parameters according to the data in the literature and the guidance of various coaching experts who use this method.

The recommended contraction interval typically ranges from 5-10 seconds. Most programmes suggest performing three to five sets of 3-5 repetitions. Load intensity is generally set at 70-90 per cent of the athlete's maximum, and 2-3 minutes of recovery are advised between sets.

These parameters may vary depending on the athlete's level of competence, professional demands, and the specific goals of the programme.